CHAPTER V

4.3(A)What sort of ethical rules should AI like self-driving car use?

An autonomous vehicle, often termed a self-driving car, possesses the ability to perceive its environment, make judgements, and operate independently of human intervention. It uses a combination of artificial intelligence, sensors, and advanced software to safely navigate highways.

A self-driving automobile employs advanced sensor technologies to observe its environment. Subsequently, advanced algorithms are employed to accurately predict the trajectory of proximate moving objects, which helps the self-driving automobile make informed decisions to navigate safely and efficiently in its environment. Self-driving cars can connect with each other via information technology, leading to enhanced coordination among the vehicles on the road. However, because automobiles are heavy and operate at high speeds, physics indicates that they possess limited manoeuvrability and sometimes cannot reach complete stops easily. Consequently, even if the communication among vehicles, sensors, and algorithms operates optimally (and surpasses existing technology), self-driving cars may not consistently have sufficient time to evade collisions with objects that rapidly alter their trajectory. According to Goodall (2014a), autonomous vehicles will occasionally clash with each other. Nonetheless, there are other dynamic elements that warrant attention. In this environment, the primary considerations are walkers, cyclists, and animals. 2015 Lin (Lin). Nevertheless, we must also consider autos operated by humans. Consequently, experts largely acknowledge that autonomous vehicles will coexist with human-driven vehicles for a longer period, resulting in mixed traffic.

The possibility exists that hazards to human life may have arisen accidentally rather than intentionally, which is an additional cause for significant concern. In the context of autonomous vehicles and the deployment of accident algorithms—specifically, the measures a car should adopt to mitigate harm in the event of an impending collision—this assertion holds considerable validity. Nyholm and Smids (2016) examine accident algorithms through the lens of the tram problem and identify three notable differences in the methodologies they employ. The ethical principles of developers, the moral and legal accountability for a self-driving

vehicle involved in an accident, and the difficulty of programming decision-making under uncertainty are notable disparities between the two. In the realm of metaethics, there is no universally acknowledged algorithm that can be deemed "right".

Recent philosophers have compared accident management in autonomous vehicles to the tram problem. Several journalists and opinion columnists have likewise accomplished this. The tram dilemma is a much-debated philosophical thought experiment involving a runaway tram, where the only means to save five individuals on the tracks is to sacrifice one person. (Thomson 1985) Various iterations of these tram scenarios differ regarding the necessity of sacrificing one individual to save five others. The most fundamental versions are purported to presage the subject of programming autonomous vehicles, particularly in how these vehicles should make ethical decisions in life-and-death situations, such as whether to sacrifice one individual to save five others. Patrick Lin states that one of the most renowned thought experiments in ethics is the tram problem. The realisation of driverless vehicles could bring this scenario closer to reality.

Judith Jarvis Thomson (1985) is notably associated with ethical discussions on moral dilemmas, especially through her analysis of the tram problem, which has profoundly influenced conversations around AI ethics and autonomous cars.
In her 1985 work, “The Trolley Problem", Thomson examines situations where causing suffering may be morally permissible to prevent greater evil. She examines the ethical justification of actively interfering (e.g., redirecting a tram) to save several lives, even at the expense of one individual's life.
Thomson distinguishes between killing and permitting death, as well as between intentional harm and foreseeable harm. She asserts that moral assessment depends not only on outcomes but also on the method of harm caused and the violation of an individual's rights.
An autonomous vehicle with a single occupant approaches a crossroads where two pedestrians and a tree are present, while a pickup truck carrying five occupants disregards a red light in its path. The operator of the autonomous car cannot prevent the crash. This incident causes the car to collide with the tree. Currently, it is unavoidable to prevent an automotive accident, and artificial intelligence has been tasked with identifying the most effective measures to mitigate probable damage, such as improving vehicle safety features and enhancing driver awareness programmes. Using the decision to collide with the pickup truck as a method of punishing people for violating traffic laws allows for a "retributivist" stance. This is due to the plausibility

of adopting this position. Thus, all five car occupants would die, and the person inside would also be in danger of death. The artificial intelligence may adopt a "selfish" strategy by opting to collide with the two pedestrians. This would kill both pedestrians but save the person in the car. It can also employ a "utilitarian" method, which involves colliding with a tree and resulting in the death of the occupant while maximising the preservation of life. This is the final, and greatest, of its skills. A person in this situation would likely be deemed "blameless", irrespective of their choice, as they would lack the time for a considered decision. This is due to their lack of opportunity to exercise free choice. Conversely, this does not apply to a self-driving automobile, as it possesses greater computational capacity than an individual. This is not the situation. Consequently, it is imperative to adopt one of the previously discussed ways in this case, but which strategy should be employed? Concerning the previously presented thought experiment, there is an additional stipulation: we possess a great deal of trust in the outcomes of each manoeuvre. This enables us to make informed decisions. An artificial intelligence can only access varying degrees of uncertainty, which does not reflect real-life conditions, leading to potential misjudgements in critical situations. The decision depicted in the initial branch, which presumed a one hundred per cent probability of fatality for the occupants of the pickup truck, would not be taken by a driving artificial intelligence designed to prioritise the minimisation of casualties. The AI would not consider the prospect of killing the individuals inside the pickup truck. In the given scenario, it was determined that there was a thirty per cent risk of their demise, whereas the likelihood of the pedestrians' demise was seventy per cent. How can this certainty be integrated into the decision-making process?

The existing work on the ethics of autonomous vehicles (AVs) concentrates on collision scenarios where an AV must determine how to allocate unavoidable harm. An unattended autonomous vehicle must either veer left, resulting in the death of five minivan occupants, or veer right, resulting in the death of a solitary motorcyclist. Such scenarios are termed "trolley instances" because they resemble a set of renowned thought exercises that have generated a substantial corpus of ethical writing and are often pejoratively labelled "trolleyology". In the initial scenario, a runaway tram is poised to strike five workers fatally; however, the operator can divert it to an alternate track, resulting in the demise of one worker instead. Moreover, as autonomous vehicles are governed by programming rather than human intuition, they must be trained to navigate such dilemmas appropriately. Alternatively, a decision of comparable moral significance must be made concerning whether the AV should remain silent about its course of action in this case. Intuitions about the conduct of autonomous vehicles have been evaluated

using trolley-based scenarios. These scenarios have been employed to evaluate concepts such as the acceptability of sacrificing a smaller group to preserve a larger one and, more contentiously, the criteria for prioritising certain individuals over others in rescue situations.
Nonetheless, the ethics of autonomous vehicles shaped by the tram conundrum have garnered much criticism. The notion that, in reality, we are seldom definite about the outcomes of our options and decisions is a fundamental aspect of reasoning.
An additional concern associated with autonomous vehicles is empirical, encompassing the topic of risk. While it is accurate that decisions pertaining to AV are taken under conditions of risk, they may also be subject to modification depending on the specific circumstances at hand. We are not simply indicating the existence of several distinct yet ambiguous choices; instead, we are alluding to the potential for an indeterminate array of options from which the autonomous vehicle (AV) may choose. Consider, for instance, the latest research in the domain of autonomous vehicle decision-making, shown by the following case:
On a dual-lane road, an autonomous vehicle (AV) is trailing the front car (FC), while a tailgater (TG) closely follows behind it. Currently, all vehicles are moving at an identical rate of 90 kilometres per hour, equivalent to highway traffic velocities. The initial occurrence in this scenario is an abrupt cessation of movement. The autonomous vehicle maintains a safe distance, enabling it to respond swiftly and halt in time to prevent a collision with the FC. Conversely, although the autonomous vehicle (AV) can accelerate sufficiently to evade a collision with the leading vehicle, the truck (TG) may lack the requisite speed to prevent colliding with the AV. The presence of oncoming traffic on the left and a barrier on the right restricts the autonomous vehicle's ability to change lanes to circumvent the obstruction. Intuitively, the autonomous vehicle (AV) appears to have two options: it can either abruptly apply the brakes, resulting in a significant rear-end collision, or it may deliberately collide with the vehicle ahead at a relatively low velocity, so diminishing the speed at which the trailing vehicle will impact the AV. Given that autonomous vehicles excel in measuring speeds and distances, is there another approach to mitigating potential harms that a human driver may lack?

Contemplate the subsequent circumstance. A self-driving vehicle with five occupants approaches a conventional automobile, such as a giant truck, which suddenly veers out of its lane and heads directly at the self-driving vehicle. The typical vehicle inexplicably veers in the opposite direction. The autonomous vehicle can quickly determine the trajectory of the approaching truck and estimate its weight. If the vehicle fails to execute a quick turn toward

the right pavement, a catastrophic collision is certain, resulting in the deaths of all five people. Regrettably, an elderly individual is walking in that vicinity, and he will be fatally injured if the autonomous vehicle executes a quick right turn and strikes him. This scenario involves human passengers in a self-driving vehicle being unable to assume control promptly. Consequently, it is imperative for the automobile to respond to the prevailing conditions. To ensure the survival of the five passengers in the self-driving car, which is probable if a head-on collision with the massive truck is averted, the vehicle must execute a manoeuvre that would likely lead to the death of one individual. It is unsurprising that circumstances like this present significant ethical dilemmas. These circumstances raise concerns about establishing priorities for the autonomous vehicle. What should those priorities be? In this scenario, should it execute a sudden manoeuvre onto the sidewalk to save the most number of individuals, or should it prioritise the innocent and collide with the oncoming truck? Should the vehicle be engineered to consistently prioritise the safety of its occupants, or should it occasionally consider alternative factors, such as equity or collective welfare, without prejudice? Without the self-driving car being programmed to respond in specific manners to ethically complex situations, such as prioritising occupant safety over collective welfare, it exhibits a significant inadequacy in its readiness to confront the realities and uncertainties of real-world traffic. This is a significant aspect to consider.

What insights into the ethics of artificial intelligence would the principles provide that could not be obtained from a fundamental ethical framework? To what degree is it possible for any framework to transform a fundamentally unethical objective into one that possesses an ethical dimension? This situation may be associated with several clear warning signs. Conversely, other concealed issues exist that were not only too intricate to foresee but also equally challenging to identify retrospectively, such as underlying psychological factors or systemic failures that contributed to the situation. These concerns are significantly more challenging to discern. Uber's decision to stop developing its driverless vehicles after a pedestrian fatality exemplifies numerous ethical infractions.

In complexity theory, the fallacy of the broken portion is considered a major and fundamental concept. Upon encountering a malfunction, the initial consideration is to identify the defective component and initiate the repair process. Which component of the engine would have malfunctioned if the airliner had crashed? Imagine an autonomous vehicle misidentified a white truck as merely a component of the sky and subsequently collided with it. In that scenario, where would the erroneous code be situated? The fractured segment is merely the most

apparent defect in this context, as is the case in numerous other situations. In numerous cases, "repairing" the impaired component will not be advantageous in preventing future issues. This is due to the systemic nature of these challenges, which frequently involve cascading failures across multiple systems, making it difficult to isolate and address individual components without considering the broader context of interdependencies. This elucidates the rationale behind this situation. Certain situations may lead to the aforementioned failures occurring "when no components are damaged, or... considered damaged." This applies in specific instances. Because we rely on linear reasoning and continuously seek to understand the rationale behind undesired impacts, we often attempt to "repair" a malfunctioning component. Our perpetual search for understanding and solutions drives this inclination.

In evaluating system safety, systems thinking requires a comprehensive investigation of the entire ecosystem, encompassing all levels of complexity. This truth is particularly applicable to the assurance of the system, as it ensures that all interactions and dependencies within the ecosystem are considered to prevent potential failures. In systems thinking, the emphasis is placed on relationships rather than individual components. This pursuit of a flawed component typically yields a superficial solution that addresses the issue without acknowledging the intricate factors behind the problem, such as the interconnectedness of various system elements and the broader context in which these issues arise. Technological advancements frequently bring up the terms "bugs" and "programming errors" in discussions. Nancy Leveson, a prominent researcher in systems engineering and safety, asserts that nearly all serious accidents involving software in the past 20 years can be attributed to flaws in requirements rather than to coding errors. Computers have been implicated in numerous serious accidents over the last two decades, which points to the urgent need for improved requirements analysis and validation in software development to prevent such flaws.

Although these flaws often manifest as code errors or user interface issues, they stem from insufficient requirements, governance, and processes. This also delineates the present condition of artificial intelligence. In discussions of artificial intelligence ethics, we often attribute responsibility to inadequate or narrowly defined training data, disregarding the pervasive ethical issues present within the organisation.

Finally, there exists a meta-philosophical quandary regarding the approach we adopt in addressing the ethics of autonomous vehicles in scenarios including risk. This is a direct consequence of the aforementioned initial two issues; however, it is not wholly derived from them. To make ethical decisions regarding autonomous vehicles (AVs), a thorough

comprehension of the philosophical issues related to them is essential. Conversely, it is essential to possess empirical insight into the conditions under which these autonomous cars would render decisions. To do this, philosophers and empirical researchers must collaborate in an unconventional manner, such as through interdisciplinary workshops and joint research projects that integrate philosophical inquiry with practical applications in autonomous vehicle design. While our previous work presents a paradigm for emulation, it fails to address broader meta-philosophical issues of how philosophers need to engage with practical and design processes, such as the integration of ethical considerations and the impact of design choices on societal outcomes.

These hazards are significant since they allow us to relax our three assumptions regarding autonomous vehicles. It is no longer essential to create a binary separation between autonomous vehicles and those controlled by humans. With adequate effort, we will be capable of acknowledging various degrees of autonomy that currently exist but are typically overlooked in debates over the ethics of autonomous cars, such as the differences between fully autonomous vehicles, semi-autonomous vehicles, and those that require human intervention. Furthermore, we can examine critical temporal dimensions of autonomous vehicle deployment, from the imminent future, where full autonomy is limited to a select few vehicles, to the plausible future, where nearly all automobiles are autonomous. Finally, we may address issues about the sorts of information necessary and the decision-making processes for cars working with diverse data, such as real-time traffic conditions, weather patterns, and user preferences, which are crucial for ensuring safe and efficient operation.

The aforementioned potential achievements in AV ethics are associated with real-world settings. The characteristics, quantity, and identity of the entities engaged in collisions—be they human, animal, or inanimate—exemplify various categories of dangers. However, there are more challenges of pressing importance that the ethics of autonomous vehicles must confront, such as the moral implications of decision-making in accident scenarios, and these issues evoke philosophical dilemmas.

A primary worry anticipated to emerge soon is how an autonomous vehicle (AV) should distribute risk in scenarios where other AVs possess differing moral obligations. Human drivers may exhibit uncertainty regarding the actions of other human drivers, which can lead to unpredictable driving behaviour and increased risk in traffic situations. Autonomous vehicles (AVs) governed by ethical algorithms may exhibit greater predictability than those operated by

humans, especially if a uniform ethical algorithm is used across all vehicles in a particular brand. Autonomous vehicles (AVs) may use this information to coordinate with other AVs and determine which vehicles to aid or hinder, contingent upon the particulars of the situation. However, this is not invariably the case.

The likelihood that a human driver has encountered a situation equal to this one is exceedingly low. Nevertheless, ethical algorithms for autonomous vehicles (AVs) may be conceptually simpler to identify, making it likely that these vehicles will increasingly encounter situations requiring such decision-making. The ethical algorithms used to govern autonomous vehicles are likely to be complex; however, as long as certain autonomous vehicles follow more "selfish" algorithms that prioritise the interests of their owners and passengers while others adhere to more "selfless" algorithms, the situation will persist unchanged.

In scenarios requiring an autonomous vehicle (AV) to collaborate with other AVs for optimal outcomes, the capacity of an AV to accurately assess the probability of other AVs conforming to specific ethical algorithms would be essential. An algorithm designed to minimise injury would prioritise collective actions by autonomous cars, such as coordinating to function as a large brake for an out-of-control truck or yielding when an autonomous vehicle must transport its driver to an emergency room. This task is beyond the capability of an individual autonomous vehicle, and attempting it would elevate the overall risk of harm for that vehicle. Consequently, it is essential for damage minimisers to convey their status as such and to determine whether other autonomous vehicles likewise adhere to harm minimisation principles. If wireless communication interconnects the AVs, we can achieve this with reasonable ease. Regardless of their status, a harm-minimising autonomous vehicle will still be motivated to gather evidence concerning the algorithms employed by other autonomous vehicles. This form of signalling presents challenges in identifying the moral commitments of individuals in the realm of autonomous vehicles (AVs) and in ensuring trust and adherence to ethical standards among these many AVs on the road, particularly when the algorithms used by different AVs may vary significantly in their ethical considerations and decision-making processes.

The modern normative ethics framework offers a diverse array of ethical models applicable to the context of autonomous vehicles (AVs). Regardless of the chosen strategy, autonomous vehicles will encounter perilous situations, albeit rarely. To resolve them, it is essential to meet at least two fundamental criteria:

a. 1. Having a gut feeling. It is imperative that it does not create a fundamental contradiction with the significant moral intuitions of potential users of AV.
2. The principles. Based on the normative equality of all human beings, it is necessary for actions to align with fundamental principles.

In arguments about the importance of intuition in ethical reasoning, the first need is more essential than in the context of autonomous vehicles (AVs). This is because autonomous vehicles represent a nascent technology that is not yet accessible to us. The introduction of autonomous vehicles guided by ethical principles that contradict moral intuitions is likely to evoke strong emotions, resulting in a diminished desire among consumers to purchase them. This would directly contradict EXT and its implied mandate for the rapid integration of autonomous vehicles into traffic. The second criterion is as crucial as the first. From a normative standpoint, all individuals are deemed equal; therefore, if autonomous vehicles were to address ethical dilemmas in a manner contradicting this principle of equality, we would possess substantial justification to dismiss the ethical framework underlying these decisions. Any form of discrimination unequivocally represents a significant violation of the principle of equality.

The autonomous vehicle needs to establish priorities. In this scenario, should it execute a sudden manoeuvre onto the sidewalk to save the most number of individuals, or should it prioritise the innocent and collide with the oncoming truck? Should the vehicle be engineered to consistently prioritise the safety of its occupants, or should it occasionally consider alternative factors, such as equity or collective welfare, without prejudice? Crucially, if the self-driving car is not engineered to respond in predefined manners to ethically charged scenarios, there exists a significant deficiency in its readiness to navigate the complexities and uncertainties of real-world traffic. This constitutes a notable exclusion. Trolley instances are obviously beneficial in the discourse surrounding these varied subjects, as they provide a framework for evaluating ethical dilemmas and decision-making processes in critical situations. To what degree are these thought experiments and the substantial literature derived from them beneficial in resolving the issue of how autonomous vehicles ought to be programmed to respond in accident scenarios where perilous collisions are either highly probable or unavoidable? Henceforth, we will contend that there are three notable areas of disanalogy that warrant the rejection of a robust parallel between the tram issue and the ethical considerations of accident algorithms for autonomous vehicles. However, from a strictly ethical

perspective, the most consequential decision-making transpires at an earlier stage. We formulate this at the planning phase, where we make decisions about programming autonomous vehicles to respond to different accident scenarios. The decisions executed by the autonomous vehicles operationalised the preceding determinations. Accident scenarios involving conventional vehicles show a higher degree of similarity to tram problems that require ethically critical decision-making. This contrasts with the mismatch between expected programming for accident scenarios involving autonomous vehicles and tram problems. For example, a driver of a conventional vehicle may abruptly encounter a situation requiring an immediate judgement on whether to swerve into one individual to avert colliding with five others. This scenario resembles a tram service more closely than the challenges faced by those who are creating accident algorithms and contingency plans for autonomous vehicles.

Furthermore, it is an unreasonable presumption to assert that an individual would be solely accountable for the decision-making regarding the programming of autonomous vehicles such as automobiles. When considering the predicament of an individual faced with a traumatic incident, we envision them as being in that condition. This is inapplicable to the context of autonomous vehicles, as these vehicles operate based on programmed algorithms and data analysis rather than human emotional responses to trauma. Numerous players execute the decision-making process for autonomous cars. The stakeholders include ordinary individuals, legal practitioners, ethicists, engineers, risk assessment specialists, automotive manufacturers, and others. These stakeholders must devise a mutually acceptable solution. The agreed-upon solution must consider the diverse interests and values that stakeholders desire to influence in the decision-making process. 8. According to John Rawls, the situation faced by the individual in the tram dilemma essentially possesses the characteristics of being generated behind a "veil of ignorance". From Rawls, 1971: The elements that can be considered are limited, and only a select handful are allowed to be included. The decision-maker may be cognisant of the presence of five individuals on the tracks and that the sole means of their salvation is to sacrifice one additional person. This resembles a trial where the jury is restricted to a very narrow range of evidence during deliberations, with only specific situational factors permissible in the decision-making process, such as the moral implications of sacrificing one person to save five others.

Veil of ignorance

Philosopher John Rawls is known for formulating the ethical concept of the veil of ignorance to ensure fairness and justice in societal interactions. A thought experiment is a technique employed to develop societal norms and structures devoid of personal bias. According to this concept, individuals are encouraged to envision themselves hidden beneath a veil of ignorance,

oblivious of their position within the social hierarchy. They remain unaware of their gender, caste, class, ethnicity, wealth, abilities, or social standing, irrespective of these variables. The deficiency of information among individuals hinders their capacity to formulate laws that are advantageous to them, as they may not fully understand how their gender, caste, class, ethnicity, wealth, abilities, or social standing impact their rights and needs.

The veil of ignorance is designed to promote impartial decision-making. Individuals are more likely to advocate for equal rights, equitable opportunities, and protections for the most vulnerable segments of society because everyone has the potential to attain wealth or fall into poverty, as well as to experience privilege or face disadvantage.

John Rawls argued that reasonable individuals, when obscured from sight, would agree on two principles of justice. Initially, it is advantageous for all to ensure equitable access to fundamental rights, including freedom of expression and legal equality. We should only accept social and economic imbalances if they benefit the most disadvantaged segments of society. Many stakeholders in the realm of ethical decision-making face a different situation. These stakeholders are collaborating to establish methods for training autonomous vehicles to respond to diverse accident scenarios, such as pedestrian crossings, sudden stops, and adverse weather conditions. Given their inability to disregard the majority of contextual and situational factors, they can only focus on a limited set of immediate situational attributes, such as the speed of surrounding vehicles, the distance to obstacles, and the behaviour of pedestrians. They can integrate all ethically significant factors they can conceive into their decision-making process for programming the autos, including considerations of safety, fairness, and the potential impact on human lives. They are capable of achieving that, and they should do so.

The essential traits of these two divergent decision-making systems are entirely dissimilar. In one situation, numerous parties conduct ethically consequential decision-making. The stakeholders are deliberating on programming a specific type of technology to respond to potential events. Furthermore, there are no limits on the types or number of elements that can be included in this decision. Conversely, in this case, the ethically pivotal decision-making is executed by an individual responding to the immediate circumstances, with only a limited number of factors considered. The primary distinction we wish to highlight is that these two decision-making scenarios differ markedly in their essential properties, particularly in the number of factors considered and the context in which the decisions are made.

Allen Wood,

A contemporary philosopher critiques John Rawls's veil of ignorance, contending that it disregards ethically significant individual variations and oversimplifies actual social realities. Wood asserts that the veil of ignorance requires individuals to disregard their true identities, values, and social connections, which significantly influence moral reasoning. He asserts that justice ought to be grounded in tangible humans, rather than abstract entities devoid of identity, and we must consider their actual social and historical situations, such as their socioeconomic status, cultural background, and personal experiences that shape their understanding of justice. He contends that moral agents need not be oblivious to their circumstances to behave justly. Instead, people can understand distinctions such as class, gender, caste, and power and still make ethical decisions through empathy, moral responsibility, and solidarity. By concealing these realities, Rawls's methodology jeopardises acknowledgement of structural inequities that necessitate deliberate recognition rather than abstraction.

Allen Wood posits that the veil of ignorance may diminish moral drive. The theory underappreciates the significance of moral commitments, collective ideals, and empathy in the pursuit of justice, as it assumes individuals act solely as detached and rational decision-makers.

Allen Wood observes that, when we disregard all other factors aside from the aforementioned considerations, there exists a significant array of factors that are, or ought to be, and in practical scenarios would be unequivocally pivotal in our moral deliberations regarding these cases, which are systematically excluded. Wood (2011, p. 70) articulates his concerns by stating that, although certain real-life decisions may result in either one or five fatalities, it is fundamentally flawed to consider the issue solely from that perspective. Wood (2011, p. 73, emphasis added) Wood observes that, when we disregard all other factors aside from the aforementioned considerations, there exists a significant array of factors that are, or ought to be, and in reality would be, fundamentally decisive in our moral deliberations regarding these cases, which are systematically overlooked. Wood (2011, p. 70) articulates his concerns by stating that, although certain real-life decisions may result in the death of either one individual or five, there is almost always an alternative outcome. Examples that may not accurately reflect reality might yet serve valuable functions in moral philosophy and other academic disciplines, such as providing frameworks for ethical decision-making in complex scenarios like the trolley problem or the programming of self-driving cars. We believe that the issue raised by Wood underscores a significant distinction between the discourse surrounding the tram dilemma and the considerations of the programming of self-driving automobiles and other autonomous vehicles in high-risk scenarios.

The crux of the matter is that, regarding the practical issue of integrating self-driving cars into actual traffic, we cannot adopt the approach of those who studied the tram problem. We cannot exclude any factors related to moral and legal accountability. The issue of how self-driving automobiles should be pre-programmed should be considered, in part, as a question of moral and legal accountability. (See Hevelke and Nida-Rümelin 2014.) We must specifically consider the accountability of those who sell or use self-driving cars, as well as the societal responsibilities associated with their presence on public roads.

In the event of severe accidents and collisions, particularly those resulting in fatalities or significant injuries, individuals tend to seek accountability from one or more parties, both ethically and legally, as they grapple with the emotional and financial repercussions of such events. This is an inescapable facet of human connection. It pertains to both conventional traffic and traffic involving autonomous vehicles. Consider a scenario in which an autonomous vehicle collides with a traditional automobile, resulting in major injuries to occupants of both vehicles, albeit without fatalities. This will undoubtedly be accompanied by legal actions. This will inevitably prompt a discussion over moral accountability for the events that transpired. If the folks involved are virtuous and rational individuals, they will likely contemplate whether the incident was attributable to their actions. We must meticulously consider and strive to achieve consensus over the moral and legal accountability of individuals in incidents involving autonomous vehicles. We must also contemplate and strive to achieve consensus over who can be held accountable for potential incidents and the resultant injuries and fatalities associated with these types of cars in traffic.

Questions about both prospective and retrospective culpability emerge in this context. Proactive measures underscore the mitigation of prospective harm. It involves responsibilities like the design of secure systems, risk assessment, the incorporation of ethical safeguards, and the continuous improvement of algorithms. In the realm of autonomous vehicles, this involves prioritising human safety, rigorously testing systems, scrutinising data for bias, and conducting ethical impact assessments to proactively prevent accidents. Backward-looking responsibility refers to the accountability individuals hold for past events, either due to their actions or their inaction. Van de Poel (2011) Both forms of responsibility are particularly pertinent to risk management and the selection of accident algorithms for autonomous vehicles. A critical group of enquiries pertains to the moral and legal accountability regarding the programming of

vehicles to navigate the diverse array of hazardous scenarios they may face in traffic. A separate set of enquiries pertains to the specific individuals who should be deemed accountable, as well as the precise nature of their responsibilities, in the event of accidents. The initial set of enquiries pertains to forward-looking responsibility, whereas the subsequent set addresses backward-looking responsibility. Both categories of enquiries are essential components of the ethics surrounding autonomous vehicles.

The responses to these challenging enquiries regarding moral and legal responsibility are complex. Instead, it is essential to confront the ethical implications of accident algorithms for autonomous vehicles, particularly how they may prioritise certain lives over others in critical situations. Discussions of the tram dilemma often disregard questions of moral and legal responsibility. For certain theoretical objectives that the trolley instances are intended to address, such actions may be entirely justified, particularly when considering the greater good or utilitarian outcomes that prioritise saving the most lives in a given scenario. Conversely, it is unjustifiable to disregard fundamental enquiries about moral and legal accountability while addressing accident algorithms for autonomous vehicles, as these considerations are crucial for ensuring ethical decision-making and public trust in the technology. We have identified a second significant disanalogy between these two subjects.

Optimising algorithms for autonomous vehicles

- **Prioritise human safety in algorithm design**. Safety limitations must be rigorously included throughout the system to ensure that the preservation of human life consistently takes precedence over objectives like speed or efficiency. This encompasses collision-avoidance systems, safe braking distances, and cautious conduct in ambiguous circumstances. Safety must be regarded as an immutable restriction within the algorithm. The system must consistently select actions that mitigate the risk of injury to pedestrians, passengers, and other road users. This entails preserving safe distances, adhering to traffic regulations, and responding prudently in ambiguous circumstances, such as when encountering unexpected obstacles or changes in traffic conditions.

Algorithms must incorporate fail-safe and fallback strategies. The vehicle should decelerate, halt safely, or relinquish control to a human operator if it experiences sensor failure,

encounters ambiguous data, or faces novel situations rather than engage in hazardous manoeuvres.

Comprehensive testing and validation are essential components of safety-focused design. Algorithms must be taught and evaluated using a variety of real-world and simulated scenarios, including infrequent and hazardous edge cases, to guarantee dependable and secure performance.

Prioritising human safety fosters public trust and ethical legitimacy. When individuals receive assurance that the algorithm prioritises human life above all else, they are more likely to embrace autonomous driving technology.

- **Embed fairness and non-discrimination rules**. Algorithms must be designed to ensure that they do not discriminate against individuals in the decision-making process based on age, gender, caste, disability, or socioeconomic status. Every human life should be assigned equal moral significance.

- Algorithms must be designed to avoid discrimination based on variables such as age, gender, caste, disability, ethnicity, or socioeconomic status. In critical situations, the system must avoid prioritising one life over another and instead focus on impartially limiting overall harm.

- To achieve this objective, developers must include ethical standards and fairness constraints in the decision-making framework. Moreover, the training data must be extensive and impartial, including individuals from many settings and backgrounds. This is essential to prevent the system from developing discriminatory patterns.

- Routine audits and monitoring algorithms are necessary to identify and rectify unfair behaviours. To ensure that the system's judgements consistently align with ethical standards, independent evaluations might be quite beneficial.
- **Ethically audited and diverse training data**. It is essential to examine training datasets for indications of bias and to confirm that they encompass a diverse range of road users,

environments, and situations. Regularly conducting data audits mitigates dangerous or discriminatory behaviours.

❖ The training data must undergo an ethical audit to identify and remove any biased, offensive, or misleading information. To achieve this, it is essential to ascertain whether specific groups, such as pedestrians, cyclists, the elderly, or individuals with disabilities, are under-represented or misrepresented in the data.

❖ To accurately represent real-world conditions, the dataset must be diverse. We must consider multiple road types, diverse traffic densities, various weather conditions, distinct lighting scenarios, and individuals from a variety of backgrounds. This capability enables the system to operate successfully across diverse surroundings and people.

❖ Continuous monitoring and updates to datasets are necessary. To maintain ethical standards, fresh data must be incorporated and re-evaluated anytime there is a modification in the social environment or road conditions.

❖ **Maintain human oversight and accountability**. If human interaction is necessary, systems must be built to facilitate it, with responsibilities explicitly designated to developers, manufacturers, and operators. The obligation for ethical conduct should not be entirely delegated to machines.

❖ It is advisable that autonomous driving systems incorporate human-in-the-loop or human-on-the-loop methodologies. These techniques would allow human operators to oversee the system's performance and intervene when it underperforms or encounters difficult and unanticipated conditions. Emergency override alternatives are essential to prevent injury.

❖ Transparent accountability frameworks must be established to determine responsibility for accidents or system failures. To prevent the dilution or evasion of duty, it is essential that manufacturers, software developers, vehicle owners, and regulators possess clearly delineated roles.

❖ Regular inspections, testing, and regulatory oversight must be conducted to ensure compliance with ethical and safety requirements. Moreover, investigators can more effectively comprehend issues and allocate responsibility equitably when the system architecture is transparent, which allows for clearer identification of potential ethical and safety violations during regular inspections and testing.

❖ **Ensure transparency and explainability**. To facilitate comprehension and discourse around algorithmic decisions, the algorithms must be meticulously built to include clear documentation of their decision-making processes and the data they use. Self-driving automotive systems must be developed with transparency and explicability to enable ethical decision-making and ensure accountability. Given that these vehicles rely on complex algorithms, it is imperative that their behaviour is understandable and open to further scrutiny, as this will help users and regulators assess the safety and ethical implications of the decisions made by these systems.

❖ To ensure transparency, it is vital to provide regulators, investigators, and other relevant authorities with clear documentation of the decision-making processes, design choices, and limits of the self-driving system. This facilitates an independent examination of the system, thereby fostering trust.

❖ In critical situations such as accidents, a self-driving car must provide explicit and comprehensible reasoning for its decision-making processes. This defines the concept of "explainability.". Thus, this enables the execution of effective investigations, fault analysis, and algorithmic enhancements.

❖ Ensuring transparency and elucidation fosters accountability, as it becomes possible to ascertain whether errors originated from data, algorithms, or the system's architecture. The assignment of blame becomes increasingly complex in the absence of justification, particularly when it is unclear which component—data, algorithms, or system architecture—contributed to the errors.

❖ **Conduct continuous testing and ethical impact assessments.** Self-driving systems should undergo thorough testing regularly in both simulated and real-world contexts to evaluate potential ethical risks and improve safety.

❖ Continuous testing and ethical impact assessments are essential to ensure that self-driving vehicles remain safe, equitable, and morally aligned throughout their lifespan. Ethical investigation cannot be a singular technique due to the constantly evolving environmental and social contexts of the real world, which require a multifaceted approach that includes stakeholder engagement, ongoing research, and adaptive policy frameworks.

❖ Continuous testing entails doing simulations and trials under real-world conditions consistently, encompassing a diverse array of environments, traffic scenarios, and atypical

edge cases. Thus, we can detect safety issues, system malfunctions, and inadvertent actions before they can cause damage.

❖ Continuous testing entails doing simulations and trials in real-world settings consistently, encompassing a diverse array of surroundings, traffic conditions, and atypical edge cases. Thus, we can detect safety issues, system malfunctions, and inadvertent actions before they can cause damage.

❖ Based on the feedback acquired via evaluations and assessments, algorithms must be revised, datasets enhanced, and ethical safeguards refined. Independent audits and regulatory assessments enhance an organization's ethical compliance.

In conclusion, autonomous cars must comply with clear and well-defined ethical standards to ensure safety, fairness, and accountability in their decision-making processes. The foremost principle is the protection of human life. An AI system must continually prioritise minimising harm while ensuring the safety of passengers, pedestrians, and other road users. Ethical programming must ensure that the vehicle complies with traffic laws, respects human rights, and eradicates discrimination in its decision-making processes. Regulation and human oversight are necessary. Governments and ethical committees ought to formulate standards and criteria to govern the functioning of autonomous vehicles in society.

Knock your brain

1 Which principle ensures that a self-driving car’s decision can be understood and explained?

a) Automation

b) Transparency

c) Efficiency

d) Autonomy

2 What should be the highest ethical priority for a self-driving car?

a) Maximizing profit for manufacturers

b) Improving travel speed

c) Ensuring human safety

d) Reducing fuel consumption

3 What ethical practice helps prevent discrimination in AI-driven vehicles?

a) Random decision-making

b) Ignoring social factors

c) Using ethically audited and diverse training data

d) Increasing automation levels

4 What does continuous testing and ethical impact assessment ensure?

a) Faster market release

b) Reduced system costs

c) Ongoing safety, fairness, and reliability

d) Elimination of human drivers

5 Which ethical principle emphasizes that AI should support, not replace, human moral judgment?

a) Full autonomy

b) Machine supremacy

c) Human-centered design

d) Algorithmic dominance

Unleash your potential

1.Should autonomous vehicles be programmed to prioritise harm reduction, even at the expense of the occupant's safety? Engage in discourse.

2.Conduct a critical assessment of the ethical challenges encountered by AI systems in road safety scenarios.